회귀분석

정재환

Data science summer institute

회귀분석
Variable Selection Procedures

정재환

Data science summer institute

Agenda

1. Introduction
2. Consequences of variable deletion
3. Criteria for evaluating equations
4. Variable selection procedures
5. Examples

Introduction

The analysis presuppose that the set of variables to be included in the equation had already been decided.

In many applications, the set of variables to be included in the equation is not predetermined, and it is often the first part of the analysis to select these variables.

Questions:

1. Which variables should be included?
2. In what form should they included, as X or transformed variables such as X^2, $\log X$?

Principle of parsimony

Try to choose a model with the smallest number of predictors (*the simplest model*) that accounts for the most substantial part of the variation in the response.

과적합[overfitting] 문제는 최근 통계학에서 주요 이슈 중의 하나이며 특히 고차원데이터[high-dimensional data] 분석의 경우 핵심 주제이다.

과적합 문제는 일반화오류[generalization error] 관점에서 고려할 수 있다. 주어진 데이터에 너무 잘 맞는 모형은, 그 변화가 있을 수 있는 새로운 데이터를 제대로 설명하지 못할 수 있으며 이는 일반화오류로 귀착된다.

모형선택 문제가 뜨거운 주제이며 범용으로 사용되는 방법론 중의 하나가 교차타당성[cross-validation]이고, 최근 널리 사용되며 연구 되는 기법으로는 라소[lasso]가 있다.

Formulation

Suppose that we have a response Y and q predictors.

We observe data $(x_{i1}, \ldots, x_{iq}, y_i), i = 1, \ldots, n$ from the model

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_q X_q + \varepsilon. \tag{1}$$

Retained predictors are denoted as $X_1, \ldots, X_p$.

Other variables $X_{p+1}, \ldots, X_q$ are omitted.

Cases

All coefficients $\beta_0, \beta_1, \ldots, \beta_q$ are nonzero so that all predictors are useful in explaining the response.

$\beta_0, \beta_1, \ldots, \beta_p$ are nonzero, but $\beta_{p+1}, \ldots, \beta_q$ are zero so that only retained variables are useful in explaining the response.

Submodel and notation

Suppose we fit a submodel

$$Y = \alpha_0 + \alpha_1 X_1 + \cdots + \alpha_p X_p + \xi \tag{2}$$

using the data $(x_{i1}, \ldots, x_{iq}, y_i)$, $i = 1, \ldots, n$, where y_i's are observed according as (1).

Denote by $\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_q$ and $\hat{y}_i$ the estimators when (1) is fitted, and by $\hat{\alpha}_0, \hat{\alpha}_1, \ldots, \hat{\alpha}_p$ and $\tilde{y}_i$ the estimators when (2) is fitted.

Unbiasedness and variance of $\hat{\alpha}_j$'s

$\hat{\alpha}_0, \hat{\alpha}_1, \ldots, \hat{\alpha}_p$ are *unbiased* for $\beta_0, \beta_1, \ldots, \beta_p$ when

- all $\beta_{p+1}, \ldots, \beta_q$ are equal to zero *OR*
- $1, X_1, \ldots, X_p$ are orthogonal to $X_{p+1}, \ldots, X_q$.

Variances for the submodel are reduced:

$$\text{var}(\hat{\alpha}_j) \le \text{var}(\hat{\beta}_j) \quad \text{and} \quad \text{var}(\tilde{y}_i) \le \text{var}(\hat{y}_i)$$

for $0 \le j \le p$ and $1 \le i \le n$.

Mean Squared Error (mse)

Mean squared error of an estimator $\hat{\theta}$ for a parameter θ is defined as $\text{mse}(\hat{\theta}) = \mathbb{E}\left(\hat{\theta} - \theta\right)^2$. Note

$$\text{mse}(\hat{\theta}) = \left(\mathbb{E}\,\hat{\theta} - \theta\right)^2 + \text{var}(\hat{\theta}) = \text{bias}^2 + \text{variance}.$$

$\hat{\beta}_j$'s are unbiased since $\mathbb{E}\,\hat{\beta}_j = \beta_j$ so that $\text{mse}(\hat{\beta}_j) = \text{var}(\hat{\beta}_j)$ for $j = 0, 1, \ldots, p$.

mse와 MSE의 차이에 주목할 필요가 있다. $\text{MSE} = \text{SSE}_p/(n-p-1)$는 Mean Square due to Error 로서 통계량[statistic]이지만 mse는 모집단 특성인 모수이다.

mse가 작은 추정량이 좋은데 *일반적으로* 편향제곱과 분산은 동시에 줄일 수 없다.

mse for $\hat{\alpha}_j$'s

$\hat{\alpha}_j$'s are biased since $\mathbb{E}\,\hat{\alpha}_j \neq \beta_j$ *in general*.

$\hat{\alpha}_j$'s *always* have less variance than $\hat{\beta}_j$ for $j = 0, 1, \ldots, p$.

Rationale for Variable Selection

The regression coefficients of the retained variables *is* estimated with smaller variance from the subset model than from the full model.

The price paid for deleting variables is in the introduction of bias in the estimators.

For variable selection, mse is a better measure for performance than variance since we need to compare possibly biased estimators.

Variable Selection in Our Setting

1. If $\mathrm{var}(\hat{\beta}_j) = \mathrm{mse}(\hat{\beta}_j) > \mathrm{mse}(\hat{\alpha}_j)$, then we include X_j since the omission of it leads to a loss of precision.
2. If $\mathrm{var}(\hat{\beta}_j) = \mathrm{mse}(\hat{\beta}_j) < \mathrm{mse}(\hat{\alpha}_j)$, then we omit X_j since the inclusion of it leads to a loss of precision.

첫째 경우는 X_j를 포함하면 편향제곱 감소량이 그 분산 증가량보다 커지므로 포함하는 것이 좋다.

둘째 경우는 X_j를 제거하면 그 분산 증가량이 편향제곱 감소량 보다 커지므로 제거하는 것이 좋다.

평가기준

mse로 부분모형의 성능을 평가할 수 있지만 실제로는 그 값을 알 수 없으므로 불가능하다.

- mse는 모수이므로 데이터만으로는 알 수 없다.

실제 부분모형과 전체모형을 비교하기 위해서는 데이터 기반 모형 평가기준이 필요하다.

서로 내포된[nested] 모형들 사이 비교는 에프검정[F-test]이 가능하지만 그렇지 않은 경우도 고려한다.

Residual Mean Square

Definition

$$\mathrm{rms}_p = \frac{\mathrm{SSE}_p}{n - p - 1} \tag{3}$$

Usage

- Between two equations, the one with the smaller rms is usually preferred, especially if the objective is forecasting.

Relation between rms and R_{ap}

Coefficient of determination

$$R_p^2 = 1 - (n - p - 1)\frac{\text{rms}_p}{\text{SST}}$$

Adjusted coefficient of determination

$$R_{ap}^2 = 1 - (n - 1)\frac{\text{rms}_p}{\text{SST}},$$

where $\text{SST} = \sum(y_i - \bar{y})^2$.

R_{ap}^2 is more appropriate than R_p^2 when comparing models with different number of predictors because R_{ap}^2 adjusts (penalizes) for the number of predictor variables in the model.

Mallows C_p

Since predicted values may be biased, we should consider mse rather than the variance.

$$J_p = \frac{1}{\sigma^2} \sum_{i=1}^{n} \mathrm{mse}(\hat{y}_i) \tag{4}$$

To estimate J_p, Mallows proposed the statistic

$$C_p = \frac{\mathrm{SSE}_p}{\hat{\sigma}^2} + 2(p+1) - n, \tag{5}$$

where $\hat{\sigma}^2$ is obtained under (1).

Usage C_p

Deviation of C_p from $p+1$ can be used as a measure of bias.

Sets of variables corresponding to points close to the line $C_p = (p+1)$ are the good or desirable subsets of variables.

Information Criteria

General formulae

$$\text{IC} = (\text{lack of fit}) + (\text{penalty for complexity of a model})$$

Usage

- Between two equations, the one with the smaller IC is usually preferred according the principle of parsimony.

Examples of Information Criteria

Akaike Information Criterion (aic) and Bayesian Information Criterion (bic) by Schwartz

$$\text{aic} = n \log(\text{SSE}_p / n) + 2(p + 1) \tag{6}$$

and

$$\text{bic} = n \log(\text{SSE}_p / n) + (p + 1) \log n. \tag{7}$$

Remarks on Information Criteria

IC's allow us to compare non-nested models.

- For comparison of a model based on X_1, X_2, X_4 with one based on (X_4, X_5), we can use aic but not an F-test.

bic gives a more parsimonious model (smaller p) than aic.

A Note on aic

Denote

$$\text{aic}_p = n \log(\text{SSE}_p/n) + 2(p+1) = \text{lof}_p + 2(p+1),$$

where lof means lack of fit.

Suppose we add one more predictor so that a new model has $\text{aic}_{p+1} = \text{lof}_{p+1} + 2(p+2)$.

Note that

$$\text{aic}_{p+1} - \text{aic}_p = \text{lof}_{p+1} - \text{lof}_p + 2$$

and

$$\text{lof}_{p+1} - \text{lof}_p \leq 0.$$

If $\text{lof}_{p+1} - \text{lof}_p > -2$, then

$$\text{aic}_{p+1} - \text{aic}_p > 0$$

so that we do not want to include the new predictor.

교차타당성 cross-validation

최근 다양한 첨단 통계 기법들이 연구되고 있음

- spline
- svm
- deep learning
- lasso

복잡모수 complexity parameter

여러 방법들의 모형선택에 사용될 수 있는 기법은 교차타당성

- 0.5 fold CV
- 4:3:3 method
- K-fold CV
- Leave-one-out CV

Uses of Regression Equations (1)

Description and Model Building

- Describe a given process or as a model for a complex system
- Typically one may want to follow principle of parsimony.

Uses of Regression Equations

Estimation and Prediction

- Predict the value of a future observation or mean response.
- Try to select a set of predictors to minimize mse.

Uses of Regression Equations

Control

- Determine the magnitude by which the value of a predictor variable must be altered to obtain a specified value of the response.
- Try to minimize the standard errors of the estimates of the coefficients.

Why do we need to select variables?

No unique 'best set' of variables.

The purpose determines the 'best set'.

- A good variable selection procedure should point out several 'adequate' subsets of variables rather than generate a single 'best' set

The process of variable selection should be viewed as an intensive analysis of

- the correlation structure of the predictors and
- how they individually and jointly affect the response under study.

How to select variables?

Different approaches to variable selection procedures are needed depending on situations.

- No multicollinearity
- Multicollinearity exists (VIFs are greater than 10).

All possible equations

Very direct and equally well to both collinear and noncollinear data.

Fit all possible subset equations.

- 2^q equations
- Pick out the three 'best' (R^2, C_p, rms_p, IC's) equations and analyze them to arrive at the final model.
- Practically infeasible when q is large.

Greedy approaches

- Forward selection procedure
- Backward elimination procedure
- Stepwise method

X_j does not have to be X_j in the model (1).

Forward Selection (FS) Procedure (I)

Pick a cutoff t_F.

1. Fit a constant.
2. Include the variable X_1 having the highest simple correlation with Y.
   2.1 If the absolute value of the t-value for X_1 is larger than t_F, it is retained; otherwise, stop the procedure.
3. Include the second variable X_2 which is defined as follows.
   2.1 Compute the residuals when Y is regressed on X_1.
   2.2 X_2 is the highest simple correlation with the residuals.
   2.3 If the absolute value of the t-value for X_2 is larger than t_F, it is retained; otherwise, stop the procedure.

Forward Selection (FS) Procedure (II)

4. Continue adding variables.
5. The procedure is terminated when the last variable entering the equation is insignificant (check with t_F).

Backward Elimination (BE) Procedure (I)

Pick a cutoff t_B.

1. Fit the full equation having q predictors.
2. Find the most insignificant variable X_1 having the smallest absolute value of the t-value among the full set or having the minimum reduction of SSE.
   2.1 If all the t-tests are significant (checked by t_B), stop; otherwise, drop X_1.
3. Fit the model with X_1 deleted.
4. Find the most insignificant variable X_2 among the remaining $(q-1)$ predictors.
   4.1 If all the t-tests are significant (checked by t_B), stop; otherwise, drop X_2.

Backward Elimination (BE) Procedure (II)

5. Continue deleting variables.
6. The procedure is terminated when all variables are significant (checked by t_B) or all variables are deleted.

Stepwise Method

A forward selection but with deletion at each stage

A variable entered in earlier stage may be eliminated at later stage.

Different levels of cutoffs

- Typically, $t_F < t_B$.
- *Guess what will happen otherwise.*

Grow and Prune Method

Growing stage

- A forward selection is adopted until we arrive at the step with a slightly overfitted model.

Pruning stage

- A backward elimination is applied to the overfitted model until we arrive at the minimal model (say, the constant model).

Best model

- Among the fitted model, we choose the best model using CV, aic or bic.

이 방법의 시작은 Breiman, Friedman, Olshen & Stone (1984) 이 제안한 Classification and Regression Tree (CART) 방법론이다. 이후 회귀스플라인[regression spline] 방법론에서 사용되고 있다.

General Remarks

An effective stopping rule

- In FS: Stop if minimum absolute t-test is less than a cutoff value 1.
- In BE: Stop if minimum absolute t-test is greater than a cutoff value 1.

The authors recommend BE over FS

- It depends on situations.
- Stepwise method is recommended when q is large.

Several equations are generated by selection procedures.

- Final model is chosen by aic or bic.
- Diagnostic should be carried out.

A Possible Strategy (I)

Examine the variables one at a time.

- Make transformation to induce symmetry and reduce skewness.

Construct pairwise scatterplots.

Fit the full linear regression model and delete insignificant variables.

- Check residuals.

교재 발간 년도를 감안하면 이러한 접근방식은 나름대로 그 의미를 가지나, 최근 설명변수 갯수가 수만개에서 수백만개에 이르는 빅데이터 시대에는 한계를 지니는 방식으로 보인다.

A Possible Strategy (II)

Examine if additional variables can be dropped.

- aic/bic would be good criteria for examining non-nested models.

For the final model, check VIF.

Attempt should be made to validate the fitted model.

Supervisor Performance Data (SPD)

Aim and goal:

1. Illustration of variable selection procedures in a non-collinear situation
2. The equation is to be constructed for understanding supervising process and the relative importance of the variables.

Multicollinearity for SPD

VIFs and pairwise correlations (Table 11.1) show that collinearity is not a problem.

$$\mathrm{vif}_1 = 2.7, \quad \mathrm{vif}_2 = 1.6, \quad \mathrm{vif}_3 = 2.3$$

$$\mathrm{vif}_4 = 3.1, \quad \mathrm{vif}_5 = 1.2, \quad \mathrm{vif}_6 = 2.0$$

Correlation Matrix for SPD (Table 11.1)

	X_1	X_2	X_3	X_4	X_5	X_6
X_1	1.000					
X_2	0.558	1.000				
X_3	0.597	0.493	1.000			
X_4	0.669	0.445	0.640	1.000		
X_5	0.188	0.147	0.116	0.377	1.000	
X_6	0.225	0.343	0.532	0.574	0.283	1.000

FS for SPD (Table 11.2)

Variables in Equation	$\min(\lvert t \rvert)$	$p+1$	aic	bic
X_1	7.74	2	118.63	121.43
$X_1 X_3$	1.57	3	118.00	122.21
$X_1 X_3 X_6$	1.29	4	118.14	123.74
$X_1 X_3 X_6 X_2$	0.59	5	119.73	126.73
$X_1 X_3 X_6 X_2 X_4$	0.47	6	121.45	129.86
$X_1 X_3 X_6 X_2 X_4 X_5$	0.26	7	123.36	133.17

Best Models from FS for SPD

Stop if $\min(|t|) < t_{0.1}(n-p-1) \Rightarrow X_1, X_3$

Stop if $\min(|t|) < 1 \Rightarrow X_1, X_3, X_6$

Here min is taken for predictors that *are not* included in the current model.

$\mathrm{aic}(118.00) \Rightarrow X_1, X_3$ whereas $\mathrm{bic}(121.43) \Rightarrow X_1$

BE for SPD (Table 11.3)

Variables in Equation	$\min(\lvert t \rvert)$	$p+1$	aic	bic
$X_1X_2X_3X_4X_5X_6$	0.26	7	123.36	133.17
$X_1X_2X_3X_4X_6$	0.47	6	121.45	129.86
$X_1X_2X_3X_6$	0.59	5	119.73	126.73
$X_1X_3X_6$	1.29	4	118.14	123.74
X_1X_3	1.57	3	118.00	122.21
X_1	7.74	2	118.63	121.43

Best Models from BE for SPD

Stop if $\min(|t|) > t_{0.5}(n-p-1) \Rightarrow X_1, X_3$

Stop if $\min(|t|) > 1 \Rightarrow X_1, X_3, X_6$

Here min is taken for predictors that *are* included in the current model.

$\mathrm{aic}(118.00) \Rightarrow X_1, X_3$ whereas $\mathrm{bic}(121.43) \Rightarrow X_1$

Collinear Data

It was pointed out that serious distortions are introduced in standard analysis with collinear data.

With a small number of collinear variables, we can evaluate all possible equations.

Possible Approaches to Variable Selection with Collinear Data

Break down the collinearity by deleting variables.

Use ridge regression.

There are many modern methods for high-dimensional data (sometimes $n \ll p$).

- LASSO (Hastie & Tibshirani)
- SCAD (Fan)
- LARS (Efron et al.)

References

Galton, F. (1886).
Regression towards mediocrity in hereditary stature.
Journal of the Anthropological Institute of Great Britain and Ireland, 15(246-263).

Pearson, K. and Lee, A. (1903).
On the laws of inheritance in man.
Biometrika, 2:357–462.